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Devices and methods for applying a hemostasis clip assembly

Abstract

A device for applying a hemostatic clip assembly includes a proximal delivery catheter having a handle assembly and an elongated catheter body defining a longitudinal axis extending distally from the handle assembly, and a distal clip assembly removably connected to a distal end of the catheter body. The distal clip assembly includes a distal clip housing, a jaw assembly and a jaw adapter yoke. The jaw assembly has a pair of jaw members fixed to the distal clip housing by a first pin oriented orthogonally relative to the longitudinal axis. The jaw adapter yoke is operatively connected to the jaw members. The proximal delivery catheter is configured to transmit linear motion along and torsion about the longitudinal axis to at least a portion of the distal clip assembly. At least one of the jaw members is configured to rotate about the first pin between an open and a closed configuration.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a U.S. National Stage application under 35 U.S.C. 371 of International Application No. PCT/US2020/058553 filed Nov. 2, 2020, which claims the benefit of priority to U.S. Provisional Patent Application Ser. No. 62/929,209 filed Nov. 1, 2019, the disclosure of which is herein incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The subject invention is directed to surgical equipment, and more particularly to hemostatic clips used endoscopic surgical procedures.

2. Description of Related Art

(2) Endoscopic or "minimally invasive" hemostatic clips are used in performance of hemostasis to stop and prevent re-bleeding, or in procedures such as ampullectomies, tissue repair and correction of other tissue defects. Such procedures are typically performed by grasping the tissue with the hemostatic clip. Benefits of using hemostatic clips in such procedures include reduced trauma to the patient, low re-bleeding rate, reduced opportunity for infection, and decreased recovery time.

(3) The subject invention provides an improved mechanism for a hemostatic clip. The novel design allows for a shorter deployed clip body, improved tissue grasping and clip locking, and an improved disconnecting feature, which are described in detail herein below, along with other novel devices and systems.

SUMMARY OF THE DISCLOSURE

(4) The subject invention is directed to a new and useful surgical device for applying a hemostatic clip assembly. The device includes a proximal delivery catheter having a proximal handle assembly and an elongated catheter body extending distally from the proximal handle assembly. The elongated catheter body defines a longitudinal axis. The device includes a distal clip assembly removably connected to a distal end of the elongated catheter body. The distal clip assembly includes a distal clip housing, a jaw assembly and a jaw adapter yoke. The jaw assembly has a pair of cooperating jaw members fixed to the distal clip housing by a first pin. The first pin is oriented orthogonally relative to the longitudinal axis. The jaw adapter yoke is operatively connected to the jaw members. The proximal delivery catheter is configured and adapted to transmit linear motion along the longitudinal axis and torsion about the longitudinal axis to at least a portion of the distal clip assembly. At least one of the jaw members is configured and adapted to rotate about the first pin between an open configuration and a closed configuration.

(5) In accordance with some embodiments, the distal clip housing includes a pair of spaced apart arms defining a slot configured and adapted to provide clearance for respective proximal portions of the jaw members to rotate relative the first pin. The distal clip assembly can include a second pin connecting between the jaw members and the jaw adapter yoke. Each jaw member can include a proximal body portion and a distal end effector. The proximal body portion of each jaw member can include a respective cam slot configured and adapted to receive the second pin and a pivot aperture configured and adapted to receive the first pin.

(6) The second pin can be configured and adapted to translate within the cam slots to move axially relative to the distal clip housing and the jaw assembly to move the jaw members between the open configuration, where respective distal tips of the jaw members are moved away from one another, the closed configuration where the respective distal tips of the jaw members are approximated towards one another to grasp tissue, and a locked configuration.

(7) In some embodiments, each cam slot defines a distal portion and a proximal portion, wherein the distal portion of each cam slot is angled relative to the proximal portion of each cam slot. The proximal portion of each cam slot can define a proximal axis extending in a first direction. The distal portion of each cam slot can define a distal axis extending at an oblique angle relative to the proximal axis, and the distal axes of each cam slot are positioned at opposite angles relative to one

another. Each cam slot can include a proximal locking neck projecting into the cam slot defining a proximal locking area. The jaw members can be in the locked configuration when the second pin is proximal relative to the proximal locking neck in the proximal locking area. The proximal locking neck can include at least one of a protrusion projecting into the cam slot or a tapered portion.
(8) The jaw adapter yoke can include a proximal receiving portion and the proximal delivery catheter includes a spring release having a distal portion configured and adapted to be received within the proximal receiving portion of the jaw adapter yoke to transmit axial and rotational force to the jaw adapter yoke. The proximal delivery catheter can include a drive wire coupled to a proximal portion of the spring release to transmit linear and rotational motion from the drive wire to the jaw adapter yoke. The proximal handle assembly can include an actuation portion coupled to a proximal end of the drive wire, and a grasping portion, wherein the actuation portion is configured and adapted to translate relative to the grasping portion to apply axial force to the drive wire.
(9) In certain embodiments, the proximal delivery catheter includes a spring tube between a proximal end of the distal clip assembly and a distal end of the catheter body. The spring tube can include at least one cantilever arm removably coupled to the distal clip housing. The at least one cantilever arm can include an inwardly extending flange that removably engages with a circumferential slot defined about a periphery of a proximal end of the distal clip housing. The proximal delivery catheter can include a spring release positioned at least partially within the spring tube. The spring tube can include an inwardly extending flange portion. The spring release can include an outwardly extending flange portion configured and adapted to interact with the inwardly extending flange portion of the spring tube to selectively deflect the at least one cantilever arm of the spring tube and release the inwardly extending flange of the at least one cantilever arm from the circumferential slot of the distal clip housing.
(10) The spring release can include a distal portion configured and adapted to be received within a receiving portion of the jaw adapter yoke to transmit linear and rotational motion to the jaw adapter yoke. The distal portion of the spring release can be divided into at least two tines. Each tine can have a mating surface selectively engageable with an inner surface of the receiving portion of the jaw adapter yoke. Each tine can be configured and adapted to deflect inwardly and release from the receiving portion when an axial force in a proximal direction is applied to the spring release.
(11) In accordance with another aspect, a device for applying a hemostatic clip assembly includes a proximal delivery catheter including a proximal handle assembly and an elongated catheter body extending distally from the proximal handle assembly. The proximal delivery catheter includes a spring tube positioned at a distal end of the elongated catheter body, a drive wire movably positioned within the elongated catheter body, and a spring release coupled to a distal end of the drive wire, the elongated catheter body defining a longitudinal axis. The spring tube includes an inwardly extending flange portion and the spring release includes an outwardly extending flange portion configured and adapted to interact with the inwardly extending flange portion of the spring tube. The device includes a distal clip assembly removably connected to a distal end of the elongated catheter body. The proximal delivery catheter is configured and adapted to transmit linear motion along the longitudinal axis and torsion about the longitudinal axis to at least a portion of the distal clip assembly.
(12) The distal clip assembly can include a distal clip housing, a jaw assembly having a pair of cooperating jaw members fixed to the distal clip housing by a first pin, and a jaw adapter yoke operatively connected to the jaw members, as previously described. A proximal body portion of each jaw member can include a respective cam slot, like cam slots described above. The distal clip housing can include a pair of spaced apart arms, like those described above. The distal clip assembly can include a second pin like that described above. Each cam slot can include a proximal locking neck projecting into the cam slot defining a proximal locking area, similar to the proximal locking neck and proximal locking area described above. Each cam slot can define a distal portion

and a proximal portion, as previously described. The proximal handle assembly can include an actuation portion and a grasping portion, as described above.
(13) In accordance with another aspect, a method for firing a hemostatic clip assembly includes positioning a distal clip assembly proximate to a target location and translating an actuation portion of a proximal handle assembly of a proximal delivery catheter relative to a grasping portion of the proximal handle assembly in at least one of a proximal direction or a distal direction. The distal clip assembly includes a distal clip housing, a jaw assembly having a pair of cooperating jaw members fixed to the distal clip housing by a first pin, and a jaw adapter yoke operatively connected to the jaw members. The proximal delivery catheter includes an elongated catheter body extending distally from the proximal handle assembly. The elongated catheter body defining a longitudinal axis. The actuation portion is operatively connected to the jaw adapter yoke via a drive wire and a spring release to transmit linear motion along the longitudinal axis and torsion about the longitudinal axis to the jaw adapter yoke. The linear motion of the jaw adapter yoke transmits the linear motion to a second pin positioned within a cam slot of at least one jaw member, thereby rotating at least one of the jaw members about the first pin between an open configuration and a closed configuration.
(14) Translating the actuation portion can include translating the actuation portion in the proximal direction to transmit the linear motion in the proximal direction to the second pin to lock the second pin behind a lock protrusion of the cam slot to lock at least one of the jaw members in a locked configuration. Translating the actuation portion can include translating the actuation portion further in the proximal direction to transmit further linear motion in the proximal direction to the spring release. The further linear motion in a proximal direction can de-couple a distal portion of the spring release from a receiving portion of the jaw adapter yoke.
(15) In some embodiments, further linear motion of the spring release in the proximal direction causing abutting between an inner diameter surface of at least one cantilever arm of a spring tube with an outwardly extending flange portion of the spring release. The spring tube can be coupled to a proximal end of the distal clip housing via the at least one cantilever arm. In certain embodiments, the abutting causes the at least one cantilever arm to deflect radially outward and disengage from the proximal end of the distal clip housing.
(16) In accordance with another aspect, a hemostatic clip assembly includes a distal clip housing defining a longitudinal axis, a jaw assembly having a pair of cooperating jaw members fixed to the distal clip housing by a first pin, and a jaw adapter yoke operatively connected to the jaw members. The jaw adapter yoke is configured and adapted to translate axially along the longitudinal axis and rotate about the longitudinal axis. At least one of the jaw members is configured and adapted to rotate about the first pin between an open configuration and a closed configuration. The first pin is oriented orthogonally relative to the longitudinal axis; and
(17) The distal clip housing can include a pair of spaced apart arms, similar to those described above. The hemostatic clip assembly can include a second pin, similar to that described above. Each jaw member and its respective cam slot can be similar to those described above.
(18) These and other features of a surgical device for applying a hemostatic clip assembly in accordance with the subject invention will become more readily apparent to those having ordinary skill in the art to which the subject invention appertains from the detailed description of the preferred embodiments taken in conjunction with the following brief description of the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) So that those skilled in the art will readily understand how to make and use the gas circulation system of the subject invention without undue experimentation, preferred embodiments thereof will

be described in detail herein below with reference to the figures wherein:
(2) FIG. **1** is a perspective view from the proximal direction of a device for applying a hemostatic clip assembly constructed in accordance with an embodiment of the present disclosure, showing a proximal delivery catheter having a proximal handle assembly and an elongated catheter body and the distal clip assembly;
(3) FIG. **2** is a perspective view of the distal clip assembly of FIG. **1**, showing a jaw assembly with a pair of cooperating jaw members fixed to the distal clip housing by a first pin;
(4) FIG. **3** is an exploded perspective view of a portion of the device of FIG. **1**, showing the showing the distal end of the proximal delivery catheter and the distal clip assembly;
(5) FIG. **4** is a perspective view of a jaw member of the device of FIG. **1**, showing the cam slot;
(6) FIG. **5** is a side elevation view of a proximal portion of the jaw member of FIG. **4**, showing proximal and distal portions of the cam slot;
(7) FIG. **6** is a perspective view of a jaw adapter yoke of the device of FIG. **1** from a proximal direction, showing the a proximal receiving portion of the jaw adapter yoke;
(8) FIG. **7** is a cross-sectional perspective view of the jaw adapter yoke of FIG. **6**, showing an inner surface of the proximal receiving portion;
(9) FIG. **8** is a perspective view of a distal clip housing of the device of FIG. **1** from a proximal direction, showing the circumferential slot defined about the periphery of a proximal end of the distal clip housing;
(10) FIG. **9** is a cross-sectional perspective view of the distal clip housing of FIG. **8**; showing a distal facing stop surface;
(11) FIG. **10** is a perspective view of a spring release of the device of FIG. **1** from a distal direction, showing a distal portion of the spring release;
(12) FIG. **11** is a cross-sectional side elevation view of the spring release of FIG. **10**; showing the mating surfaces of the tines of the distal portion of the spring release;
(13) FIG. **12** is a perspective view of a spring tube of the device of FIG. **1** from a distal direction, showing cantilever arms of the spring tube;
(14) FIG. **13** is a cross-sectional perspective view of the spring tube of FIG. **12**, showing inwardly extending flanges of the cantilever arms;
(15) FIG. **14** is a cross-sectional side elevation view of a portion of the device of FIG. **1**, showing the jaw members in the open configuration where respective distal tips of the jaw members are moved away from one another to grasp a target area of tissue;
(16) FIG. **15** is a cross-sectional side elevation view of a portion of the device of FIG. **1**, showing the jaw members in the closed configuration where the respective distal tips of the jaw members are approximated towards one another to grasp tissue;
(17) FIG. **16** is a cross-sectional side elevation view of a portion of the device of FIG. **1**, showing the jaw members in the locked configuration where the second pin is proximal relative to a proximal locking neck;
(18) FIG. **17** is a cross-sectional side elevation detail view of a portion of the device of FIG. **1**, showing the engagement between the spring release and the jaw adapter yoke and between the spring tube and the distal clip housing before firing;
(19) FIG. **18** is a cross-sectional side elevation detail view of a portion of the device of FIG. **1**, showing the release of the spring release from the jaw adapter yoke and the release of the spring tube from the distal clip housing when firing the device;
(20) FIG. **19** is a cross-sectional side elevation detail view of a portion of the device of FIG. **1**, schematically showing the inward deflection of the tines of the spring release as they move proximally relative to the jaw adapter yoke when firing the device;
(21) FIG. **20** is a cross-sectional side elevation detail view of a portion of the device of FIG. **1**, schematically showing the outward deflection of the cantilever arm of the spring tube as the spring release moves proximally relative to the spring tube when firing the device;

(22) FIG. **21** is a cross-sectional side elevation view of a portion of the device of FIG. **1**, showing the spring tube and spring release disconnected from the distal clip assembly for removal of the proximal delivery catheter;
(23) FIG. **22** is a perspective view of a portion of a device for applying a hemostatic clip assembly constructed in accordance with another embodiment of the present disclosure, schematically showing the distal clip housing is rotatable along with the spring release;
(24) FIG. **23** is a cross-sectional perspective view of the distal clip housing of FIG. **22**, showing the distal clip housing including a diametrical center bar to engage with tines of the spring release for rotation'
(25) FIG. **24** is a perspective view of portions of another embodiment of a release pin and another embodiment of a jaw adapter yoke constructed in accordance with the present disclosure, showing the jaw adapter yoke having flat bosses extending from a proximal end to engage with flat outer surfaces on tines of the spring release;
(26) FIG. **25** is a perspective view of another embodiment of a jaw member for use with the device for applying a hemostatic clip assembly of FIG. **1**, showing a cam slot;
(27) FIG. **26** is a side elevation view of the jaw member of FIG. **25**, showing two protrusions in the cam slot;
(28) FIG. **27** is a side elevation view of another embodiment of a jaw member for use with the device for applying a hemostatic clip assembly of FIG. **1**, showing a tapered portion and a lip in the cam slot;
(29) FIG. **28** is a side elevation view of another embodiment of a jaw member for use with the device for applying a hemostatic clip assembly of FIG. **1**, showing a slot in the jaw member; and
(30) FIG. **29** is a side elevation view of another embodiment of a jaw member for use with the device for applying a hemostatic clip assembly of FIG. **1**, showing a reverse slope in the proximal portion of the cam slot.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(31) Referring now to the drawings wherein like reference numerals identify similar structural elements and features of the subject invention, there is illustrated in FIG. **1** a surgical device for applying a hemostatic clip assembly in a patient, and more particularly, for separating the hemostatic clip assembly to function as a short-term implant constructed in accordance with a preferred embodiment of the subject disclosure and is designated generally by reference numeral **10**.
(32) As shown in FIGS. **1-2**, a surgical device **10** for applying a hemostatic clip assembly **100** includes proximal delivery catheter **101** and the distal clip assembly **100**. The distal clip assembly **100**, e.g., a hemostasis clip, separates from the delivery catheter **101** to function as a short-term implant to stop and prevent re-bleeding, or in procedures such as ampullectomies, tissue repair and correction of other tissue defects. Such procedures are typically performed by grasping the tissue with the hemostatic clip. Using hemostatic clips in such procedures can result in benefits such as reduced trauma to the patient, low re-bleeding rate, reduced opportunity for infection, and decreased recovery time.
(33) With continued reference to FIGS. **1-2**, the proximal delivery catheter **101** has a proximal handle assembly **103** and an elongated catheter body **105** extending distally from the proximal handle assembly **103**. The elongated catheter body **105** defines a longitudinal axis A. The proximal handle assembly **103** includes an actuation portion **115** (e.g., actuator, as labeled in FIG. **2**) coupled to a proximal end **111** of the drive wire **109**, and a grasping portion **107**. The actuation portion **115** is configured and adapted to translate relative to the grasping portion **107** to apply an axial force to the drive wire **109**. Grasping portion **107** and actuation portion **115** are configured and adapted to rotate relative to a cap **170** and catheter body **105**, thereby also rotating drive wire **109**. Internal annular slots on the distal portion of grasping portion **107** interact with annular tabs on inside diameter of end cap **170** to prevent axial motion of actuation portion **115** and grasping portion but

allow rotation.
(34) With continued reference to FIGS. **1-2**, the proximal delivery catheter **101** includes a spring tube **142** between a proximal end of the distal clip assembly **100** and a distal end **143** of the catheter body **105**. A proximal end **144** of the spring tube **142** is coupled to the distal end **143** of the catheter body via weld, adhesive, or other means. The distal clip assembly **100** includes a distal clip housing **102** and a jaw assembly **104** pivotally connected to the distal clip housing **102**. The jaw assembly **104** has a pair of cooperating jaw members **108** fixed to the distal clip housing **102** by a first pin **110**. The first pin **110** is oriented orthogonally relative to the longitudinal axis A. The spring tube **142** includes cantilever arms **146** configured and adapted to be removably coupled to the distal clip housing **102**, described in more detail below. The hemostatic clip assembly **100** is removably connected to a distal end **143** of the elongated catheter body **105** via the spring tube **142**. The proximal delivery catheter **101** is configured and adapted to transmit linear motion along the longitudinal axis A and torsion about the longitudinal axis A to at least a portion of the distal clip assembly **100**.
(35) With reference now to FIGS. **2-3**, the distal clip assembly **100** includes a jaw adapter yoke **106**. The proximal delivery catheter **101** includes a spring release **136** having a distal portion **138** configured and adapted to be received within the jaw adapter yoke **106**. The distal clip assembly **100** includes a second pin **118** connecting between the jaw members **108** and the jaw adapter yoke **106**. The jaw members **108** are configured and adapted to rotate about the first pin **110** between an open configuration and a closed configuration. Each jaw member **108** includes a proximal body portion **116** and a distal end effector **120**. The proximal body portion **116** of each jaw member **108** includes a respective cam slot **122** configured and adapted to receive the second pin **118**. Jaw members **108** are driven by the second pin **118**, e.g., a cam pin, moving along the cam slots **122** of the jaw members **108**. The second pin **118** is configured and adapted to translate within the cam slots to move axially relative to the distal clip housing **102** and the jaw assembly **104** to move the jaw members **108** between the open configuration where respective distal tips **124** of the jaw members **108** are moved away from one another, the closed configuration where the respective distal tips **124** of the jaw members **108** are approximated towards one another to grasp tissue, and a locked configuration.
(36) With reference now to FIGS. **3-5**, each jaw member **108** includes a pivot aperture **134** configured and adapted to receive the first pin **110**. Each jaw member **108** of the jaw assembly **104** is identical to the other member **108**, allowing additional economy of scale. The distal end effectors **120** of each jaw member **108** can include at least one pointed peak, multiple peaks, of different or similar size at their distal tips **124**. Distal end effectors **120** could also terminate in a combination of pointed peaks and rounded peaks to balance tissue pressure, allowing jaw members **108** to hook tissue with at least one peak and provide atraumatic contact with at least one peak. As shown in FIG. **14**, the tooth (or teeth, peaks, etc.) may create an angle ω relative to an axis J of their respective jaw arms **108** between zero and 180 degrees, optimizing the approach angle of distal tips **124** relative to tissue surface. In the embodiment of FIG. **14**, the angle ω of a tip axis T relative to axis J is approximately 90 degrees. It is contemplated, however, that the angle ω could be at 0 degrees, such that the tip simply extends from axis J, it could be at 45 degrees, or 180 degrees, where the tip is hooked around such that the tip axis T direction is parallel to axis J. The angle and design of jaw members **108** will be optimized for single jaw tissue retention force during manipulation or tissue apposition. The distance between the pivot aperture **134** and the cam slot **122** dictate the moment arm that translates axial translation to jaw rotation/actuation.
(37) As shown in FIGS. **4-5**, each cam slot **122** defines a distal portion **126** and a proximal portion **128**, wherein the distal portion **126** of each cam slot **122** is angled relative to the proximal portion **128** of each cam slot **122**. The proximal portion **128** of each cam slot **122** defines a proximal axis P extending in a first direction. The distal portion **126** of each cam slot **122** defines a distal axis D extending at an oblique angle θ relative to the proximal axis P, and the distal axes D of each cam

slot **122** are positioned at opposite angles relative to one another, as shown in FIGS. **14-16**. The angle of a respective distal axis D relative to proximal axis P can be fine-tuned to provide optimal tissue clamping force given a user's maximum acceptable input force.
(38) With continued reference to FIGS. **4-5**, Each cam slot **122** includes a proximal locking neck **130**, e.g., a locking feature, projecting into the cam slot **122** defining a proximal locking area **132**. The jaw members **108** are in the locked configuration when the second pin **118** is proximal relative to the proximal locking neck **130** in the proximal locking area **132**. The proximal locking neck **130** includes a protrusion **131** projecting into the cam slot **122**. Lock protrusion **131**, e.g., a detent, creates a narrowing of cam slot **122** to form the proximal locking neck **130** that interferes with the outer diameter of the second pin **118** as it moves axially in the proximal direction. The continued axial translation of pin **118** forces a widening of the cam slot **122** in an elastic manner and creates an additional resistance force on the internal drivetrain, e.g., spring release **136** and spring tube **142**. Once the second pin **118** crests the inflection point on the protrusion **131**, it will snap into place behind the protrusion **131**, effectively locking the jaws in a closed position. The shape of lock protrusion can vary and can be an arcuate, triangular, or slanted feature. Lock protrusion **131** may also be achieved by reversing the slope of cam slot **122** such that it inflects passed the 0 degree orientation with respect to the axis A of the catheter, described in more detail below. Various embodiments for the proximal locking neck are described below in FIGS. **25-29**.
(39) As shown in FIGS. **3** and **6-7**, the jaw adapter yoke **106** includes a pin aperture **160** and is operatively connected to the jaw members **108** via second pin **118**. The jaw adapter yoke **106** is circular component with two arms **113** extending towards the distal end of the yoke **106** that form a slot **161** therebetween. The slot **161** allows the proximal portions **116** of the jaw members **108** rotate around first pin **110**. An aperture **160** is formed in each arm **113** and is in a transverse direction to a longitudinal axis of the jaw adapter yoke and the longitudinal axis A of the catheter body **105**. The apertures **160** receive the second pin **118** and can be assembled using orbital riveting or laser tack welding. The jaw adapter yoke **106** includes a proximal receiving portion **133** and slides linearly inside of distal clip housing **102** to drive second pin **118** along the cam slots **122**. The proximal receiving portion **133** of the jaw adapter yoke **106** has a square and/or rectangular shape with flat surfaces **153** to mate with a snap feature **141** (described below) on the distal portion **138** of spring release **136**, allowing linear force transmission up to a predetermined value.
(40) In certain embodiments, it is contemplated that flat features could be machined onto the back proximal face of the jaw adapter yoke **106**, i.e. a transverse slot bisecting the yoke to form bosses that interface with flat faces on the tines of spring release **136**, which is shown by embodiments of spring release **736** and yoke **706**, shown in FIG. **24** and described below. In some embodiments, the proximal receiving portion **133**, e.g., the rectangular hole, could be circular in cross section where an arcuate neck surface **157** on spring release **136** could be in an interference fit with the circular receiving portion such that friction and radial loading would transmit torque between spring release **136** jaw adapter yoke **106**.
(41) As shown in FIGS. **3** and **8-9**, the distal clip housing **102** includes a pair of spaced apart arms **112** defining a slot **114** configured and adapted to provide clearance for respective proximal portions **116** of the jaw members **108** to rotate relative the first pin **110**. The distal clip housing **102** connects via a snap fit connection to the spring tube **142** via a circumferential slot **148**, e.g., a stepped retention ring, defined about a periphery of a proximal end **151** of the distal clip housing **102**. The distal clip housing **102** includes a tapered outer diameter portion **149** proximal relative to the circumferential slot **148**. During assembly of the distal clip assembly **100** to the proximal delivery catheter **101**, the tapered outer diameter portion **149** pushes the cantilever arms **146** of the spring tube **142** radially outward allowing the 90 degree bent tabs (e.g., the inwardly extending flanges **158** described in more detail below) to seat in the circumferential slot **148** of distal clip housing **102**. The distal clip housing **102** includes a transverse hole **147** oriented perpendicular to the longitudinal axis A to accept the first pin **110**, e.g., the pivot pin, which couples to jaw members

108. An inner surface **165** of distal clip housing **102** further includes a distal facing stop surface **163**. The inner diameter surface **165** of distal clip housing **102** allows for axial transmission of jaw adapter yoke **106**, until jaw adapter yoke **106** contacts the hard stop created by distal facing stop surface **163**.

(42) With reference to FIGS. **3**, **10-11** and **14**, the distal portion **138** of the spring release **136** is configured and adapted to be received within the proximal receiving portion **133** of the jaw adapter yoke **106**, shown in FIGS. **6-7**, to transmit axial and rotational force to the jaw adapter yoke **106**. The rotation of drive wire **109** about the longitudinal axis A drives rotation of spring release **136**, jaw adaptor yoke **106**, distal clip housing **102** and jaw assembly **104** about the longitudinal axis A relative to catheter body **105** and spring tube **142**. The distal portion of the spring release **136** is divided into at least two tines **139** and includes a snap feature **141** at the distal most tip of each tine **139**. The snap feature **141** includes a tapered outer diameter surface **145** at the distal tip of each tine **139** and a mating surface **156** selectively engageable with an inner surface **135** of the receiving portion **133** of the jaw adapter yoke **106**. Each tine includes flat outer surfaces **155** on each side to engage with the square/rectangular inner surfaces **153** of receiving portion **133** to allow torque transmission from drive wire **109** to the distal clip assembly **100**. The drive wire **109** is mechanically coupled to a proximal portion **140** of the spring release **136** to transmit linear and rotational motion from the drive wire **109** to the jaw adapter yoke **106**. The proximal portion **140** of the spring release **136** being the portion proximal relative to tines **139**. The spring release **136** includes a receiving bore **162** opening in the proximal direction for receiving and coupling the drive wire **109** to the spring release **136**. The spring release **136** includes an outwardly extending flange portion **152** and an outwardly projecting stop flange **154** positioned proximal to the outwardly extending flange portion **152**.

(43) As shown in FIGS. **2-3** and **12-13**, the cantilever arms **146** of the spring tube **142** can be laser cut or stamped from the tube body, creating at least one cantilever arm **146**. Each cantilever arm **146** includes an inwardly extending flange **158** that removably engages with a circumferential slot **148** defined about a periphery of a proximal end **151** of the distal clip housing **102**. The inwardly extending flange **158** is a 90-degree tab bent inwards that mates with corresponding circumferential slot **148** of jaw housing. The spring release **136** is positioned at least partially within the spring tube **142**. The spring tube **142** includes inwardly extending v-shaped flange portion **150**, e.g., a first formed feature, and inner stop flanges **137**, e.g., stop tabs.

(44) With reference now to FIGS. **14-17**, some of the various configurations of device **10** are shown. In Fig. the In FIG. **15**, the device **10** is in a closed configuration and the second pin **118** is translated in a more proximal position within each cam slot **122** relative to the open configuration, but second pin **118** is still distal of the locking neck **130** and the protrusion **131**. In the closed configuration, the respective distal tips **124** of the jaw members **108** are approximated towards one another to grasp tissue **15** (but not necessarily in abutment with one another). In FIGS. **16-17**, the device **10** is in a locked configuration and the second pin **118** is in a proximal position relative to the locking necks **130** and their respective protrusions **131**. In the locked configuration, the second pin **118** is within the proximal locking area **132** of each cam slot **122**.

(45) As shown in FIGS. **17-20**, once the second pin **118** is in the proximal locking area **132**, further axial movement of the spring release **136** in a proximal direction (e.g. away from the tissue **15**) acts to “fire” the distal clip assembly **100** by releasing the distal clip assembly **100** from the proximal delivery catheter **101**. The further linear motion of the spring release **136** in the proximal direction puts the spring release **136** in tension against jaw adapter yoke **106** due to abutment between mating surface **156** of the snap feature **141** and the inner surface **135** of the receiving portion **133**. This tension causes each tine **139** to act as a spring and deflect inwardly, shown schematically by the inwardly pointing arrows in FIG. **19**, and release from the receiving portion **133**. The release force required to detach spring release **136** from the adapter yoke **106** can be tuned by adjusting the length C of each tine **139**, the thickness T of each tine **139**, and/or the angle σ of mating surface

relative to the longitudinal axis A, show in FIG. **17**. The angle σ of mating surface can range from 30 to 60 degrees, preferably being 45 degrees. The ratio of length C to thickness T can range from 8:1 to 10:1, e.g., **9**:**1**. These dimensions provide the desired elastic behavior to ensure consistent release.

(46) With continued reference to FIGS. **17-21**, as the spring release **136** moves proximally relative to the jaw adapter yoke **106**, it also moves proximally relative to the spring tube **142**, thereby causing abutment between an inner diameter surface of the v-shaped flange portions **150** of the cantilever arms **146** of a spring tube **142** with an outwardly extending flange portion **152** of the spring release **136**. The abutting causes each cantilever arm **146** to deflect radially outward and disengage the inwardly extending flanges **158** from the circumferential slot **148** at the proximal end **151** of the distal clip housing **102**. The outwardly projecting stop flange **154** is configured and adapted to engage with the inner stop flange **137** of spring tube **142** to stop spring release **136** from receding proximally into the catheter body **105**. Full disengagement (e.g. "firing") of the distal clip assembly **100** is realized through both the inward deflection of the tines **139** of spring release **136** and the outward deflection of the cantilever arms **146** of spring tube **142**.

(47) With continued reference to FIGS. **17-21**, a single spring component, spring release **136**, disengages two springs (tines **139** and cantilever arms **146**) simultaneously generating an improved disconnect mechanism that enhances the ability to reposition the clip assembly **100** prior to deployment by simplifying the feedback to the user into a single tactile signal. It also makes accidental deployment of the clip assembly **100** less likely, as fewer components are used to realize disengagement. Because there are fewer components, less space is needed in the distal clip assembly **100**, allowing for a shorter clip body. The shorter clip "stem" or overall length of deployed clip relative to jaw size is seen as an improvement. Additionally, the firing mechanism is elastic, and permanent deformation, e.g. breakage, is not required to deploy the clip assembly **100**. As such, deployment can be reversible in some embodiments to a certain extent, which allows for the possibility of a multi-use delivery catheter. As shown in FIG. **21**, after firing, proximal delivery catheter **101** can then be removed from the surgical site, leaving the distal clip assembly **100** to function as a short-term implant.

(48) A method for firing a hemostatic clip assembly, e.g. distal clip assembly **100**, includes positioning the distal clip assembly proximate to a target location, e.g. near tissue **15** as shown in FIG. **14**, and translating an actuation portion, e.g. actuation portion **115**, of a proximal handle assembly, e.g. proximal handle assembly **103**, of a proximal delivery catheter, e.g. proximal delivery catheter **101**, relative to a grasping portion, e.g. grasping portion **107**, of the proximal handle assembly in at least one of a proximal direction or a distal direction. The actuation portion is operatively connected to a jaw adapter yoke, e.g. jaw adapter yoke **106**, via a drive wire, e.g. drive wire **109**, and a spring release, e.g. spring release **136** to transmit linear motion to the jaw adapter yoke. The linear motion of the jaw adapter yoke transmits the linear motion to a second pin, e.g. second pin **118**, positioned within a cam slot, e.g. cam slot **122**, of at least one jaw member, e.g. jaw members **108**, thereby rotating at least one of the jaw members about the first pin between an open configuration and a closed configuration.

(49) The method includes translating the actuation portion in the proximal direction to transmit the linear motion in the proximal direction to the second pin, as shown in FIG. **15**, to lock the second pin, as shown behind a lock protrusion, e.g. lock protrusion **131**, of the cam slot to lock at least one of the jaw members in a locked configuration, as shown in FIG. **16**. Translating the actuation portion includes translating the actuation portion further in the proximal direction to transmit further linear motion in the proximal direction to the spring release, as shown in FIG. **18**. The further linear motion in a proximal direction de-coupling a distal portion, e.g. distal portion **138**, of the spring release from a receiving portion, e.g., receiving portion **133**, of the jaw adapter yoke, as shown in FIG. **21**.

(50) Referring now to FIGS. **25-29**, several different embodiments for the jaw members are

described. In FIG. **25-26**, an embodiment of a jaw member **208** is shown. Jaw member **208** is similar to jaw members **108** in that it can be used in the jaw assembly **104** and the distal clip assembly **100**. Jaw member **208** also includes a distal end effector **220** similar to distal end effector **120**. The main difference between jaw member **208** and jaw member **108** is that jaw member **208** includes a cam slot **222** in a proximal portion **216** of the jaw member **208** where the cam slot **222** includes a proximal locking neck **230** with two protrusions **231** projecting into the cam slot **222** defining a proximal locking area **232**. Protrusions **231**, e.g., detents, interfere with the outer diameter of the of a cam pin, e.g., pin **118**. For jaw member **208**, the continued axial translation of cam pin **118** forces a widening of the cam slot **222** in an elastic manner and creates an additional resistance force on the internal drivetrain, e.g., spring release **136** and spring tube **142**. Once the cam pin **118** crests the inflection point on the protrusions **231**, it will snap into place behind the protrusion **231**, effectively locking the jaws in a closed position. Because a drive wire, e.g., drive wire **109**, operatively connected to jaw member **208** can only transmit limited compression, a user will not be able to translate sufficient force from a handle assembly, e.g., handle assembly **103**, distally to move cam pin **118** out of locking area **132** relative to the protrusions **231** to “unlock” the cam pin.

(51) As shown in FIG. **27**, another embodiment of a jaw member **308** is shown. Jaw member **308** is similar to jaw members **108** in that it can be used in the jaw assembly **104** and the distal clip assembly **100**. Jaw member **308** also includes a distal end effector similar to distal end effector **120**. The main difference between jaw member **308** and jaw member **108** is that jaw member **308** includes a cam slot **322** in a proximal portion **316** of the jaw member **308** having a locking neck **330** formed by a tapered portion **331**, e.g., a triangular ramp, having a lip **333**. A proximal locking area **332**, similar to locking area **132**, is defined by the locking neck **330** proximally from the lip **333**. This geometry allows an easier transmission of axial force to normal force on the internal walls of cam slot **322**, requiring less force to initiate locking. The lip **333** positioned distally relative to the proximal locking area **332** will also prevent axial movement of a cam pin, e.g., cam pin **118**, after locking is achieved.

(52) With reference now to FIG. **28**, another embodiment of a jaw member **408** is shown. Jaw member **408** is similar to jaw members **108** in that it can be used in the jaw assembly **104** and the distal clip assembly **100**. Jaw member **408** also includes a distal end effector similar to distal end effector **120**. The main difference between jaw member **408** and jaw member **108** is that jaw member **408** includes a cam slot **422** in a proximal portion **416** of the jaw member **408** having a locking neck **430** formed by a tapered portion **431**, e.g., a triangular ramp, terminating in a slot **433**. This open contour creates a cantilever lock arm **435** on the bottom wall of cam slot **422**. This results in a decreased force required to lock the clip, and results in a higher rate of successful locking in instances where the jaw members **408** are not perfectly parallel to each other, as deflection in the cantilever lock arm **435** can accommodate some axis offset of the jaw members **108**. A proximal locking area **432**, similar to locking area **132**, is defined by the locking neck **430** and positioned proximally from the tapered portion **431**.

(53) As shown in FIG. **29**, another embodiment of a jaw member **508** is shown. Jaw member **508** is similar to jaw member **108** in that it can be used in the jaw assembly **104** and the distal clip assembly **100**. Jaw member **508** also includes a distal end effector similar to distal end effector **120**. The main difference between jaw member **508** and jaw member **108** is that jaw member **508** includes a cam slot **522** in a proximal portion **516** of the jaw member **508** having a locking neck **530** formed by a slope reversal on a proximal portion **528** of the cam slot **522**. In other words, instead of a proximal axis P of proximal portion **528** being parallel to a longitudinal axis A of a catheter body, e.g., catheter body **105**, proximal axis P is angled radially outward relative to axis A resulting in a locking force due to cantilever deflection. In this instance, the user will feel a gradual increase in feedback force, and then a sudden decrease. Once a cam pin, e.g., second pin **118**, has crested an inflection point **529** of the pin track (again, relative to the longitudinal axis of the clip

body, which is parallel to longitudinal axis A of catheter body at rest) the slope direction changes and begins to force the clip open ever so slightly (0-10 degrees of angulation between jaws. Subsequent unlocking of the jaw members **508** would require equal distal movement of the cam pin relative to a pivot pin, e.g., first pin **110**, which is prevented by the spring force required to pass the cam pin over the inflection point during distal translation. Again, an elongate drive wire, e.g., drive wire **109**, will not be able to transmit sufficient compressive force to actuate the cam pin distally, effectively locking the clip. The cam slot **522** of jaw member **508** has the as the added benefit of accommodating some amount of tissue thickness between the jaw members **508** without incurring bending stress in the jaw members **508**.

(54) As shown in FIGS. **22-23**, an alternate mechanism for torque transmission in device **10** is proposed where a distal clip housing **602** includes a diametrical center bar **670** extended across opening **671** generating two slots **671***a* and **671***b* on either side with two flat surfaces **670***a* and **670***b* on the inside of the through hole. These parallel flat surfaces **670***a* and **670***b* contact the internal flat edges **168** of tines **139** of spring release **136**, shown in FIG. **10**, that are formed from a slot between tines **139**. The remaining portions of distal clip housing **602** are the same as distal clip housing **102**. Distal clip housing **602** can be used in lieu of distal clip housing **102** in device **10**. Moreover, when using distal clip housing **602** in device **10**, jaw adapter yoke **106** can have a circular proximal receiving portion **133** (instead of rectangular/square), as the torque does not need to be transmitted via the flats **155** on the spring release **136** to the receiving portion **133**.

(55) As shown in FIG. **24**, another alternate mechanism for torque transmission in device **10** is proposed through the connection between alternative embodiments of a spring release **736** and a jaw adapter yoke **706**. Spring release **736** is the same as spring release **136** except instead of an arcuate outer surface like that on tines **139**, tines **739** include flat portions **770** to abut bosses **771** having flat surfaces **768** extending from the proximal end of the jaw adapter yoke **706** that interface with the aforementioned flat portions **770** of tines **739**. Jaw adapter yoke **706** is the same as jaw adapter yoke **106** except for the bosses **771**. Spring release **736** and jaw adapter yoke **706** can be used in device **10** in lieu of spring release **136** and jaw adapter yoke **106**.

(56) The methods and systems of the present disclosure, as described above and shown in the drawings, provide for a surgical device with superior properties including simplified user feedback, reduced accidental deployment of the clip assembly and a shorter clip body. Additionally, the firing mechanism is elastic, and permanent deformation, e.g., breakage, is not required to deploy the clip assembly. While the apparatus and methods of the subject disclosure have been showing and described with reference to embodiments, those skilled in the art will readily appreciate that changes and/or modifications may be made thereto without departing from the spirit and score of the subject disclosure.

Claims

1. A device for applying a hemostatic clip assembly, the device comprising: a proximal delivery catheter including a proximal handle assembly and an elongated catheter body extending distally from the proximal handle assembly, the elongated catheter body defining a longitudinal axis; and a distal clip assembly removably connected to a distal end of the elongated catheter body, the distal clip assembly including a distal clip housing, a jaw assembly having a pair of cooperating jaw members fixed to the distal clip housing by a first pin, the first pin oriented orthogonally relative to the longitudinal axis, and a jaw adapter yoke operatively connected to the jaw members, wherein the proximal delivery catheter is configured and adapted to transmit linear motion along the longitudinal axis and torsion about the longitudinal axis to at least a portion of the distal clip assembly, wherein at least one of the jaw members is configured and adapted to rotate about the first pin between an open configuration and a closed configuration wherein the distal clip assembly includes a second pin connecting between the jaw members and the jaw adapter yoke, wherein each

jaw member includes a proximal body portion and a distal end effector, wherein the proximal body portion of each jaw member includes a respective cam slot configured and adapted to receive the second pin and a pivot aperture configured and adapted to receive the first pin, wherein the second pin is configured and adapted to translate within the cam slots to move axially relative to the distal clip housing and the jaw assembly to move the jaw members between the open configuration where respective distal tips of the jaw members are moved away from one another, the closed configuration where the respective distal tips of the jaw members are approximated towards one another to grasp tissue, and a locked configuration, and wherein each cam slot includes a proximal locking neck projecting into the cam slot defining a proximal locking area, wherein the jaw members are in the locked configuration when the second pin is proximal relative to the proximal locking neck in the proximal locking area.

2. The device as recited in claim 1, wherein the distal clip housing includes a pair of spaced apart arms defining a slot configured and adapted to provide clearance for respective proximal portions of the jaw members to rotate relative the first pin.

3. The device as recited in claim 1, wherein each cam slot defines a distal portion and a proximal portion, wherein the distal portion of each cam slot is angled relative to the proximal portion of each cam slot.

4. The device as recited in claim 3, wherein the proximal portion of each cam slot defines a proximal axis extending in a first direction, the distal portion of each cam slot defines a distal axis extending at an oblique angle relative to the proximal axis, and the distal axes of each cam slot are positioned at opposite angles relative to one another.

5. The device as recited in claim 1, wherein the proximal locking neck includes at least one of a protrusion projecting into the cam slot or a tapered portion.

6. A device for applying a hemostatic clip assembly, the device comprising: a proximal delivery catheter including a proximal handle assembly and an elongated catheter body extending distally from the proximal handle assembly, the elongated catheter body defining a longitudinal axis; and a distal clip assembly removably connected to a distal end of the elongated catheter body. the distal clip assembly including a distal clip housing, a jaw assembly having a pair of cooperating jaw members fixed to the distal clip housing by a first pin, the first pin oriented orthogonally relative to the longitudinal axis, and a jaw adapter yoke operatively connected to the jaw members, wherein the proximal delivery catheter is configured and adapted to transmit linear motion along the longitudinal axis and torsion about the longitudinal axis to at least a portion of the distal clip assembly, wherein at least one of the jaw members is configured and adapted to rotate about the first pin between an open configuration and a closed configuration, wherein the jaw adapter yoke includes a proximal receiving portion and the proximal delivery catheter includes a spring release having a distal portion configured and adapted to be received within the proximal receiving portion of the jaw adapter yoke to transmit axial and rotational force to the jaw adapter yoke.

7. The device as recited in claim 6, wherein the proximal delivery catheter includes a drive wire coupled to a proximal portion of the spring release to transmit linear and rotational motion from the drive wire to the jaw adapter yoke.

8. The device as recited in claim 7, wherein the proximal handle assembly includes an actuation portion coupled to a proximal end of the drive wire, and a grasping portion, wherein the actuation portion is configured and adapted to translate relative to the grasping portion to apply axial force to the drive wire.

9. The device as recited in claim 6, wherein the distal clip housing includes a pair of spaced apart arms defining a slot configured and adapted to provide clearance for respective proximal portions of the jaw members to rotate relative the first pin.

10. A device for applying a hemostatic clip assembly, the device comprising: a proximal delivery catheter including a proximal handle assembly and an elongated catheter body extending distally from the proximal handle assembly, the elongated catheter body defining a longitudinal axis; and a

distal clip assembly removably connected to a distal end of the elongated catheter body, the distal clip assembly including a distal clip housing, a jaw assembly having a pair of cooperating jaw members fixed to the distal clip housing by a first pin, the first pin oriented orthogonally relative to the longitudinal axis, and a jaw adapter yoke operatively connected to the jaw members, wherein the proximal delivery catheter is configured and adapted to transmit linear motion along the longitudinal axis and torsion about the longitudinal axis to at least a portion of the distal clip assembly, wherein at least one of the jaw members is configured and adapted to rotate about the first pin between an open configuration and a closed configuration, wherein the proximal delivery catheter includes a spring tube between a proximal end of the distal clip assembly and a distal end of the catheter body, wherein the spring tube includes at least one cantilever arm removably coupled to the distal clip housing, wherein the at least one cantilever arm includes an inwardly extending flange that removably engages with a circumferential slot defined about a periphery of a proximal end of the distal clip housing, and wherein the proximal delivery catheter includes a spring release positioned at least partially within the spring tube, wherein the spring tube includes an inwardly extending flange portion, wherein the spring release includes an outwardly extending flange portion configured and adapted to interact with the inwardly extending flange portion of the spring tube to selectively deflect the at least one cantilever arm of the spring tube and release the inwardly extending flange of the at least one cantilever arm from the circumferential slot of the distal clip housing.

11. The device as recited in claim 10, wherein the spring release includes a distal portion configured and adapted to be received within a receiving portion of the jaw adapter yoke to transmit linear and rotational motion to the jaw adapter yoke.

12. The device as recited in claim 11, wherein the distal portion of the spring release is divided into at least two tines, wherein each tine has a mating surface selectively engageable with an inner surface of the receiving portion of the jaw adapter yoke.

13. The device as recited in claim 12, wherein each tine is configured and adapted to deflect inwardly and release from the receiving portion when an axial force in a proximal direction is applied to the spring release.

14. The device as recited in claim 10, wherein the distal clip housing includes a pair of spaced apart arms defining a slot configured and adapted to provide clearance for respective proximal portions of the jaw members to rotate relative the first pin.